

Supplementary Materials for paper:

HBHS-DF : A Novel Hardness-based Hybrid Sampling Deep Forest for Imbalanced Drug-Target Interaction Prediction

S1. Pseudocode

Algorithm 1 describes the training process of our HBHS-DF. It should be noted that in Step 3, the oversampled dataset $D_{oversampling}$ is not involved in the equalization under-sampling processes of EUS-R and EUS-C. Once equalization under-sampling is completed, $D_{oversampling}$ is merged with the under-sampled result to train the base learner, thereby enhancing the model's generalization ability.

Algorithm 1 HBHS-DF Algorithm

Input: Training Set D

Parameter: sampling rate K , accuracy threshold $accuracy_threshold$, max layer L , early stop rounds R

Output: HBHS-DF classifier C

1. $C \leftarrow \{\}, D_{oversampling} \leftarrow \{\}, layer_{best} \leftarrow 1$
 2. **for** $l \leftarrow 1$ to L **do**
 3. Train EUS-Rs and EUS-Cs using parameter K under 5-fold cross-validation on D and $D_{oversampling}$
 4. Append EUS-Rs and EUS-Cs to C
 5. Concatenate the original feature vector and augmented feature vector
 6. Calculate instance hardness TD and identify difficult samples with parameter $accuracy_threshold$
 7. $D_{oversampling} \leftarrow$ perform oversampling on difficult samples with parameter K
 8. Evaluate the performance of layer l
 9. **If** performance of layer l is better than $layer_{best}$ **do**
 10. $layer_{best} \leftarrow l$
 11. **end if**
 12. **if** $l - layer_{best} > R$ **do**
 13. Break
 14. **end if**
 15. **end for**
 16. **return** C
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S2. Hyperparameter Configurations of HBHS-DF

Hyperparameters	Description	Value					
K	Under-sampling rate and oversampling rate in HBHS-DF.	dataset	IR=5	IR=10	IR=15	IR=20	IR=25
		BioSNAP	3	3	5	5	5
		Human	2	2	3	3	3
		Drugbank	2	3	4	5	5
$Accuracy_threshold$	The threshold is used to determine hard samples for oversampling. The value of the left boundary of the first interval where the accuracy rate falls below $Accuracy_threshold$ is used to identify hard samples.	0.70					
max_layer	The maximum number of layers of the HBHS-DF model.	2					
$Stride_{TD}$	The intervals are divided based on the TD value with $Stride_{TD}$ as the step size, thereby determining the hard samples.	0.05					

S3. Information about the Employed Evaluation Metrics

As mentioned in the main text, we employed $AUPR$, $F1$, ACC and AUC for performance comparison. (1-6) illustrate the calculation process of these for metrics. TP/FP indicates the positive predictions are right/wrong, and TN/FN indicates the negative predictions are right/wrong. REC indicates the accuracy of the classifier's prediction of positive samples, and PRE indicates the proportion of positive samples in the positive prediction. $F1$ is a measure that considers both PRE and REC , maintaining a reasonable measurement on imbalanced datasets. AUC is a performance index to measure the quality of classifiers, where $\sum I(P_{pos}, P_{neg})$ represents the number of positive samples with higher prediction probability than negative samples, and N_{pos}/N_{neg} represents the number of positive/negative samples. It is not affected by the classification threshold and reflects the overall prediction ability of a model. $AUPR$ measures the performance of a binary classification model at different thresholds, which calculates the average precision at different recall levels, and we use the trapezoidal rule to estimate the area under the curve.

$$ACC = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$REC = \frac{TP}{TP + FP} \quad (2)$$

$$PRE = \frac{TP}{TP + FN} \quad (3)$$

$$F1 = \frac{2 * PRE * REC}{PRE + REC} \quad (4)$$

$$AUC = \frac{\sum I(P_{pos}, P_{neg})}{N_{pos} * N_{neg}} \quad (5)$$

$$AUPR = \sum_{k=1}^{N_s} (REC_k - REC_{k-1}) * PRE_k \quad (6)$$

S4. Performance comparison on benchmark datasets on metrics

TABLE I. PERFROMANCE COMPARISON ON BRNCHMARK DATASETS ON $F1$

Dataset		RF	CatBoost	SPE	EASE	DF	BINDTI	DrugBAN	HBHS-DF
BioSNAP	IR=5	0.830±0.004	0.822±0.006	0.859±0.004	0.788±0.007	0.882±0.004	0.805±0.010	0.801±0.017	0.889±0.003
	IR=10	0.809±0.004	0.803±0.004	0.842±0.004	0.745±0.014	0.874±0.003	0.729±0.023	0.755±0.019	0.880±0.003
	IR=15	0.793±0.004	0.790±0.004	0.826±0.004	0.691±0.013	0.864±0.005	0.690±0.014	0.703±0.020	0.871±0.003
	IR=20	0.781±0.004	0.779±0.003	0.813±0.004	0.648±0.013	0.859±0.003	0.660±0.023	0.684±0.017	0.866±0.002
	IR=25	0.770±0.005	0.770±0.005	0.802±0.007	0.629±0.017	0.852±0.004	0.616±0.015	0.659±0.024	0.862±0.004
Human	IR=5	0.873±0.016	0.887±0.014	0.905±0.007	0.861±0.007	0.896±0.007	0.858±0.017	0.841±0.014	0.914±0.005
	IR=10	0.855±0.005	0.869±0.009	0.888±0.010	0.825±0.010	0.878±0.007	0.784±0.048	0.802±0.013	0.897±0.007
	IR=15	0.848±0.010	0.860±0.006	0.876±0.006	0.792±0.013	0.870±0.010	0.737±0.009	0.749±0.021	0.886±0.002
	IR=20	0.846±0.007	0.854±0.007	0.869±0.004	0.771±0.008	0.866±0.006	0.718±0.024	0.719±0.017	0.883±0.007
	IR=25	0.842±0.009	0.848±0.009	0.860±0.009	0.747±0.013	0.860±0.009	0.682±0.015	0.672±0.020	0.874±0.006
DrugBank	IR=5	0.835±0.002	0.820±0.004	0.861±0.002	0.778±0.005	0.887±0.003	0.811±0.008	0.816±0.011	0.895±0.001
	IR=10	0.811±0.005	0.803±0.002	0.845±0.002	0.720±0.012	0.880±0.001	0.761±0.015	0.768±0.021	0.888±0.002
	IR=15	0.796±0.004	0.790±0.005	0.832±0.003	0.674±0.015	0.871±0.002	0.706±0.020	0.734±0.021	0.881±0.001
	IR=20	0.785±0.006	0.782±0.001	0.819±0.003	0.638±0.009	0.867±0.004	0.674±0.025	0.702±0.012	0.874±0.003
	IR=25	0.775±0.003	0.775±0.001	0.807±0.003	0.623±0.013	0.861±0.002	0.649±0.017	0.668±0.014	0.870±0.002
Average Rank (R_j)		4.51	4.57	2.76	7.17	2.25	7.11	6.63	1.00

TABLE II. PERFROMANCE COMPARISON ON BRNCHMARK DATASETS ON ACC

Dataset		RF	CatBoost	SPE	EASE	DF	BINDTI	DrugBAN	HBHS-DF
BioSNAP	IR=5	0.921±0.002	0.916±0.002	0.923±0.001	0.864±0.006	0.940±0.002	0.873±0.010	0.870±0.016	0.941±0.001
	IR=10	0.950±0.001	0.948±0.001	0.947±0.001	0.885±0.009	0.963±0.001	0.868±0.019	0.886±0.015	0.962±0.001
	IR=15	0.963±0.001	0.962±0.001	0.957±0.001	0.878±0.010	0.972±0.001	0.873±0.011	0.881±0.016	0.972±0.001
	IR=20	0.970±0.001	0.970±0.000	0.963±0.001	0.869±0.011	0.978±0.001	0.874±0.020	0.892±0.013	0.977±0.000
	IR=25	0.975±0.001	0.974±0.001	0.967±0.002	0.873±0.016	0.981±0.001	0.856±0.014	0.892±0.019	0.981±0.001
Human	IR=5	0.938±0.008	0.943±0.007	0.949±0.004	0.916±0.005	0.946±0.004	0.912±0.013	0.899±0.011	0.953±0.004
	IR=10	0.960±0.003	0.963±0.003	0.964±0.004	0.932±0.005	0.963±0.003	0.899±0.034	0.914±0.008	0.966±0.003
	IR=15	0.971±0.002	0.972±0.002	0.971±0.001	0.937±0.006	0.973±0.002	0.900±0.007	0.908±0.013	0.974±0.001
	IR=20	0.977±0.001	0.978±0.001	0.976±0.001	0.942±0.003	0.979±0.001	0.909±0.016	0.911±0.011	0.979±0.002
	IR=25	0.981±0.001	0.981±0.001	0.979±0.001	0.943±0.005	0.982±0.001	0.903±0.009	0.895±0.016	0.982±0.001
DrugBank	IR=5	0.924±0.002	0.915±0.002	0.924±0.001	0.854±0.005	0.942±0.002	0.879±0.007	0.883±0.009	0.944±0.001
	IR=10	0.952±0.001	0.948±0.001	0.948±0.001	0.865±0.011	0.965±0.000	0.891±0.011	0.896±0.015	0.965±0.001
	IR=15	0.964±0.001	0.962±0.001	0.959±0.001	0.864±0.014	0.974±0.001	0.885±0.016	0.903±0.014	0.974±0.001
	IR=20	0.971±0.000	0.970±0.000	0.964±0.001	0.860±0.008	0.979±0.000	0.885±0.019	0.904±0.008	0.979±0.000
	IR=25	0.976±0.000	0.975±0.000	0.968±0.001	0.868±0.013	0.982±0.000	0.887±0.013	0.899±0.010	0.982±0.000
Average Rank (R_j)		3.55	3.97	4.28	7.03	1.67	7.24	6.73	1.53

TABLE III. PERFORMANCE COMPARISON ON BRNCHMARK DATASETS ON *AUC*

Dataset		RF	CatBoost	SPE	EASE	DF	BINDTI	DrugBAN	HBHS-DF
BioSNAP	IR=5	0.937±0.004	0.921±0.004	0.930±0.003	0.908±0.004	0.941±0.003	0.928±0.004	0.925±0.004	0.952±0.002
	IR=10	0.938±0.004	0.922±0.006	0.932±0.004	0.909±0.005	0.939±0.004	0.920±0.007	0.929±0.006	0.954±0.003
	IR=15	0.937±0.004	0.922±0.003	0.931±0.001	0.907±0.005	0.940±0.003	0.925±0.002	0.929±0.002	0.956±0.002
	IR=20	0.937±0.001	0.921±0.002	0.932±0.003	0.906±0.003	0.940±0.003	0.921±0.005	0.931±0.004	0.956±0.002
	IR=25	0.937±0.003	0.922±0.003	0.933±0.004	0.906±0.004	0.940±0.001	0.924±0.005	0.932±0.003	0.956±0.002
Human	IR=5	0.955±0.003	0.967±0.005	0.962±0.006	0.947±0.007	0.953±0.003	0.957±0.007	0.949±0.006	0.966±0.004
	IR=10	0.951±0.005	0.965±0.004	0.959±0.003	0.942±0.008	0.949±0.004	0.956±0.008	0.951±0.007	0.967±0.003
	IR=15	0.950±0.003	0.963±0.005	0.960±0.005	0.942±0.006	0.950±0.008	0.958±0.009	0.956±0.009	0.966±0.006
	IR=20	0.948±0.005	0.962±0.006	0.958±0.006	0.940±0.007	0.949±0.006	0.958±0.009	0.954±0.003	0.967±0.004
	IR=25	0.948±0.003	0.964±0.004	0.958±0.004	0.939±0.007	0.950±0.006	0.961±0.002	0.954±0.007	0.965±0.004
DrugBank	IR=5	0.938±0.005	0.918±0.005	0.929±0.003	0.906±0.003	0.940±0.004	0.921±0.007	0.923±0.006	0.951±0.002
	IR=10	0.939±0.003	0.920±0.002	0.929±0.003	0.906±0.006	0.940±0.002	0.924±0.005	0.927±0.005	0.954±0.003
	IR=15	0.939±0.000	0.921±0.002	0.931±0.002	0.905±0.006	0.940±0.003	0.922±0.007	0.929±0.007	0.956±0.001
	IR=20	0.937±0.002	0.920±0.003	0.929±0.002	0.906±0.004	0.938±0.004	0.924±0.006	0.929±0.007	0.955±0.002
	IR=25	0.938±0.002	0.920±0.003	0.932±0.003	0.905±0.006	0.938±0.004	0.919±0.003	0.930±0.007	0.956±0.002
Average Rank (R_j)		3.89	5.07	4.20	7.81	3.51	5.33	4.99	1.20

S5. Details of the Friedman test and Nemenyi test

About the Friedman test, (7) shows the original Friedman statistics. And the improved version is described in (8), which is distributed according to the F -distribution with $(K - 1)$ and $(K - 1)(N - 1)$ degrees of freedom. The experiments were conducted under five-fold cross validation with five various IRs across three benchmark datasets with HBHS-DF and seven algorithms. Therefore, N and K are set to 75 and 8, respectively.

$$X_F^2 = \frac{12N}{K(K+1)} \left[\sum_j R_j^2 - \frac{K(K+1)^2}{4} \right] \quad (7)$$

$$F_F = \frac{(N-1)X_F^2}{K(K-1) - X_F^2} \quad (8)$$

$$R_j = \frac{1}{N} \sum_i r_i^j, i \in [1, N] \quad (9)$$

where R_j is the average rank of each algorithm across the datasets.

The Nemenyi test is conducted to verify the performance differences between pairwise algorithms. The Critical Difference (CD) can be calculated as:

$$CD = q_\alpha \sqrt{\frac{K(K+1)}{6N}} \quad (10)$$

S6. Instance Hardness-based Oversampling Analysis

This section examines the specific impact of the proposed strategy on the prediction performance for difficult samples. As TD serves as the indicator of instance hardness, we analyzed the distribution of quantity and prediction accuracy of samples across different TD levels. Specifically, we divided TD into intervals and conducted statistics on the minority class samples in the test set on the first and last layers. Fig. 1 shows the statistical results on the BioSNAP dataset with IR=5.

The bars in Fig. 1 depict the number of samples within each sample interval, while the lines illustrate the prediction accuracy of the corresponding samples. Compared with the first layer, in the last layer, there are more samples in the intervals with lower TD . Besides, when the TD value is greater than 0.40, prediction accuracy of the last layer is significantly higher than that of the first layer. Therefore, it can be concluded that the oversampling strategy effectively improves the model's generalization capability.

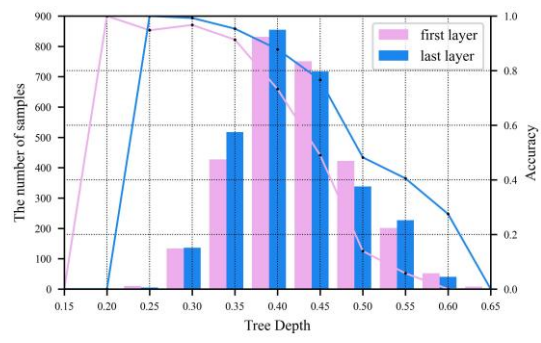


Fig. 1. The quantity and accuracy distribution of samples over tree deepth.